QFI and method of moments: $N = 20,\ B/J = 0.40$

QFI, F_h — $\mathcal{M}_h[\hat{S}_z]$ — $\mathcal{M}_h[\hat{X}]$ — $\mathcal{M}_h[\hat{Y}]$, $\hat{Y} = i\hat{S}_z\hat{X}$

(a) $\left(\left|E_1\right\rangle + \left|E_{\overline{1}}\right\rangle\right)/\sqrt{2}$

(b) $\left|E_1\right\rangle$

(c) $\left|\uparrow\cdots\uparrow\right\rangle$

(d) $\left(\left|\uparrow\cdots\uparrow\right\rangle + \left|\downarrow\cdots\downarrow\right\rangle\right)/\sqrt{2}$

$\frac{2\pi}{\Delta_{3\overline{3}}}$ $\frac{2\pi}{\Delta_{2\overline{2}}}$ $\frac{2\pi}{\Delta_{1\overline{1}}}$

$\mathcal{M}_h/(Nt)^2$, $F_h/(Nt)^2$,

0.00 0.08 0.16 0.24 0.32

10^0 10^1 10^2 10^3 10^4 10^5 10^6

t/T

$\mathcal{M}_h[\hat{O}] = (\partial_h\langle\hat{O}\rangle)^2/\mathrm{Var}(\hat{O})$